



slackr :: CHEAT SHEET

Send files, messages, R objects, and images to Slack directly from R

Installation

CRAN version
`install.packages("slackr")`

Development version
`devtools::install_github("mrkaye97/slackr")`

Setup

- Go to <https://api.slack.com/apps>
- Click **"Create New App"** and then follow the setup instructions

Multi-channel bot

- Click **"OAuth & Permissions"** under **"Features"**
- Enable the following scopes in order to get all of the functionality:
 - `channels:read`
 - `chat:write`
 - `users:read`
 - `chat:write.customize`
 - `files:read`
 - `chat:write.public`
 - `groups:read`
 - `im:write`
 - `groups:write`
 - `incoming-webhook`
 - `channels:history`
- Click **"Install to Workspace"**
- Select a channel (for webhook messages)
- Copy the Bot User OAuth Access Token
- Click **"Incoming Webhooks"** under **"Features"**
- Copy the Webhook URL

Single-channel bot

- Click **"Incoming Webhooks"** under **"Features"**
- Turn the **"Activate Incoming Webhooks"** switch on
- Click **"Add New Webhook to Workspace"**
- Select the channel you'd like the bot to post to
- Copy the Webhook URL

Setup environment variables for Slack API access

```
slackr_setup(  
  channel = "#general",  
  username = "slackr",  
  icon_emoji = "",  
  incoming_webhook_url = "",  
  bot_user_oauth_token = "",  
  config_file = "~/.slackr",  
  echo = FALSE,  
  cacheChannels = TRUE,  
  cache_dir = ""  
)
```

Generate Config Text (.txt) File (optional)

```
bot_user_oauth_token: xoxb-...  
channel: #yourchannel  
username: yourusername  
incoming_webhook_url: https://  
hooks.slack.com/services/XXXXX/XXXXX/XXXXX
```

Common Functions

Post a ggplot to a Slack channel

```
ggslackr(  
  plot = last_plot(),  
  channels = Sys.getenv("SLACK_CHANNEL"),  
  scale = 1,  
  width = par("din")[1],  
  height = par("din")[2],  
  units = c("in", "cm", "mm"),  
  dpi = 300,  
  limitsize = TRUE,  
  bot_user_oauth_token =  
    Sys.getenv("SLACK_BOT_USER_OAUTH_TOKEN"),  
  file = "ggplot",  
  ...  
)
```

Save R objects to an RData file on Slack

```
save_slackr(  
  ...,  
  channels = Sys.getenv("SLACK_CHANNEL"),  
  file = "slackr",  
  bot_user_oauth_token =  
    Sys.getenv("SLACK_BOT_USER_OAUTH_TOKEN"),  
  plot_text = ""  
)
```

Send result of R expressions to a Slack channel via webhook API

```
slackr_bot(  
  ...,  
  channel = "",  
  username = "",  
  icon_emoji = "",  
  incoming_webhook_url =  
    Sys.getenv("SLACK_INCOMING_URL_PREFIX")  
)
```

Delete the specified number of messages from the channel

```
slackr_delete(  
  count,  
  channel = Sys.getenv("SLACK_CHANNEL"),  
  bot_user_oauth_token =  
    Sys.getenv("SLACK_BOT_USER_OAUTH_TOKEN")  
)
```

Send a message to a Slack channel

```
slackr_msg(  
  txt = "",  
  channel = Sys.getenv("SLACK_CHANNEL"),  
  username = Sys.getenv("SLACK_USERNAME"),  
  icon_emoji = Sys.getenv("SLACK_ICON_EMOJI"),  
  bot_user_oauth_token =  
    Sys.getenv("SLACK_BOT_USER_OAUTH_TOKEN"),  
  ...  
)
```

Send a file to Slack

```
slackr_upload(  
  filename,  
  title = basename(filename),  
  initial_comment = basename(filename),  
  channels = Sys.getenv("SLACK_CHANNEL"),  
  bot_user_oauth_token =  
    Sys.getenv("SLACK_BOT_USER_OAUTH_TOKEN")  
)
```

Other Usages

<code>auth_test()</code>	Checks authentication & identity against the Slack API
<code>call_slack_api()</code>	A wrapper function to call the Slack API with authentication and pagination
<code>convert_response_to_tibble()</code>	Convert Slack API json response to tibble
<code>register_onexit()</code>	Append text_slackr as on.exit to functions
<code>slackr_census_fun()</code>	Create a cache of the users and channels in the workspace in order to limit API requests
<code>slackr_channels()</code>	Get a data frame of Slack channels
<code>slackr_chtrans()</code>	Translate vector of channel names to channel IDs for API
<code>slackr_dev()</code>	Send the graphics contents of the current device to a Slack channel
<code>slackr_history()</code>	Reads history of a channel
<code>slackr_ims()</code>	Get a data frame of Slack IM IDs
<code>slackr_users()</code>	Get a data frame of Slack users
<code>text_slackr()</code>	Sends basic text to a slack channel
<code>tex_slackr()</code>	Post a tex output to a Slack channel
<code>with_pagination()</code>	Calls the slack API with pagination using cursors

Vignettes

Creating a single-channel bot

`vignette('webhook-setup', package = 'slackr')`

Creating a fully-functional multi-channel bot

`vignette('scoped-bot-setup', package = 'slackr')`

Usage

`vignette('using-slackr', package = 'slackr')`